MILITARY SPECIFICATION

CARTRIDGE, 7.62MM, MATCH, XM852

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers Cartridge, 7.62MM, Match, XM852 intended for marksmanship training and competitive firing in 7.62MM National Match Rifles.

2. APPLICABLE DOCUMENTS

2.1 Issue of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILTIARY

MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification For.

STANDARDS

FEDERAL

MIL-STD-105	-	Sampling Procedures and Tables for Inspections of Attributes.
MIL-STD-109	-	Quality Assurance Terms and Definitions.
MIL-STD-636	-	Visual Inspection Standards for Small Arms Ammunition Through Caliber .50.
MIL-STD-644	-	Visual Inspection Standards and Inspection Procedures for Inspection of Packaging, Packing and Marking of Small Arms Ammunition.
MIL-STD-1168	-	Lot Numbering of Ammunition.
MIL-STD-1170	-	Visual Standards and Comparison Methods for Evaluating Grain Configuration in 7.62MM Cartridge Cases.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Federal Test Method Standard No. 151 - Metals, Test Methods.

2.2 Other publications.

DEPARTMENT OF DEFENSE

SCATP-7.62MM	-	Ammunition Ballistic Acceptance Test Methods.

DRAWINGS

9341551	-	Cartridge, 7.62MM, Match, XM852.
7643674	-	Classification of Cartridge Case Defects Small Arms Ammunition.
9341748	-	Label, Carton.
9341749	-	Packing and Marking Cartridges, 7.62MM, Match, XM852, Cartons, Box, Ammunition, M2A1.
9341750	-	Packaging and Marking, Cartridges 7.62MM, Match, XM852: Cartons; box ammunition, M2A1; box wirebound
LI 9341551	-	Index of Inspection Equipment List for: Cartridge, 7.62MM, Match, XM852.

(Copies of specifications, stnadards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article inspection. Requirements for the submission of first article samples shall be as specified in the contract. Unless otherwise specified, the first article shall include the pilot pack.

3.2 Material. Materials shall be in accordance with the applicable drawings and specifications.

3.3 Part. The part shall comply with all requirements specified on Dwg. 9341551, all associated drawings, and with all requirements specified in applicable specifications.

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3.4 Bullet extraction. The force necessary to extract the bullet from the cartridge case shall be not less than 20 pounds.

3.5 Residual stress. The cartridge case shall not split when subjected to one percent mercurous nitrate solution for 15 minutes.

3.6 Accuracy. The average extreme spread of all targets, each target containing 14 shots fired at 600 yards, shall not exceed 8.0 inches.

3.7 Velocity. The average velocity of the sample cartridge, conditioned at 70° ± 2° Fahrenheit (F), shall be 2,550 feet per second (ft/sec) plus or minus 30 ft/sec, at 78 feet from the muzzle of the weapon. The standard deviation of the velocities shall not exceed 28 ft/sec.

3.8 Chamber pressure. The average chamber pressure of the sample cartridges conditioned at 70°F ± 2°F, shall not exceed 50,000 pounds per square inch (psi). Neither the chamber pressure of an individual sample cartridge, nor the average chamber pressure of the sample cartridges plus three standard deviations of chamber pressure shall not exceed 55,000 psi.

3.9 Port pressure. The average port pressure of the sample cartridges, conditioned at 70°F ± 2°F, shall be 12,500 psi ± 2,000 psi.

3.10 Function and casualty. The cartridge shall function without casualty.

3.11 Grain configuration. The grain configuration of the sidewall of the finished cartridge case shall fall within the range defined by the grain configuration standards illustrated in MIL-STD-1170, Figures 1 through 6.

3.12 Workmanship. The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.12.1 Metal defects. The cartridge shall be free of folds, wrinkles, deep draw scratches, scaly metal, dents and other defects.

3.12.2 Foreign matter. The cartridge shall be free of corrosion, strains, discoloration, dirt, oil, and smears of lacquer.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item.

a. First Article Inspection.

b. Quality Conformance Inspection.

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

PART DESCRIPTION	DRAWINGS	QUANTITY
Cartridge, 7.62MM Match, XM852	9341551	400

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I herein.

4.3.3 Rejection. See MIL-A-48078.

4.3.3.1 Function and casualty.

a. Any misfires.

b. Any bullet remaining in bore.

c. Two or more primer leaks due to perforation in firing pin indent in primer cup.

d. Two or more primer leaks due to escape of gas through primer cup other than "C" above.

e. Four or more primer leaks due to escape of gas around primer cup.

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f. Two or more primer leaks due to primer that remains in pocket, but is physically loose.

g. Any blown primer.

h. Any primer that falls out of pocket on retraction of bolt.

i. Six or more longitudinal splits at neck and shoulder (I and S).

j. Two or more longitudinal splits, body (J).

k. Any longitudinal split, body (K).

l. Any longitudinal split to head (L).

m. Any longitudinal split through head (M).

n. Two or more circumferential ruptures.

o. Any circumferential rupture: partial body (K).

p. Any circumferential rupture: partial head (L).

q. Any circumferential rupture: complete.

r. Any failure to extract.

s. Any weapon stoppage.

4.4 Quality conformance inspection. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, each inspection lot shall contain:

a. Primer from one lot and one manufacturer.

b. Propellant from one lot and one manufacturer.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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TABLE I. FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63450(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Cartridge, 7.62MM, Match, XM852				9341551
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY PARAGRAPH REFERENCE /INSPECTION METHOD
	Cartridge, 7.62MM, Match, XM852		ACC/ REJ		
	Dwg. 9341551				
	Examination for defects	20	0/1	3.3	4.4.2.1
	Bullet extraction	20	0/1	3.4	4.5.1
	Residual stress	20	0/1	3.5	4.5.2
	Grain configuration	10	0/1	3.11	4.5.8
	Hardness	10	0/1	3.3	4.5.9
	Accuracy	84	0/1	3.6	4.5.3
	Velocity	20	0/1	3.7	4.5.4
	Chamber pressure	20	0/1	3.8	4.5.5
	Port pressure	20	0/1	3.9	4.5.6
	Function & Casualty				
	Rifle, 7.62MM, M14 (National Match)	100	See Para 4.3.3.1	3.10	4.5.7

NOTES:

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TABLE II. QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63450 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 5		DRAWING NUMBER
4.4.2.1	Cartridge, 7.62MM, Match, XM852				9341551
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:					
1	Weight, min. 1/		100%	3.3	Gage
2	Split case in K, L, or M location 2/		100%	3.3	Gage
3	Perforated Case 2/		100%	3.3	Visual
Major					
101	Cartridge, corroded or stained, if etched 2/		.25	3.3	Visual
102	Case, round head 2/		.25	3.3	Visual
103	Split case in I, S, or J location 2/		.25	3.3	Visual
104	Case, beveled underside of head 2/		.25	3.3	Visual
105	No chamfer on head (rim) 2/		.25	3.3	Visual
106	No visible evidence of mouth anneal 2/		.25	3.3	Visual
107	Split bullet jacket 2/		.25	3.3	Visual
108	Loose bullet 2/		.25	3.3	Visual
109	No primer 2/		.25	3.3	Visual
110	Cocked primer 2/		.25	3.3	Visual
111	Inverted primer 2/		.25	3.3	Visual
112	Loose primer 2/		.25	3.3	Visual
113	Total length, incorrect		.25	3.3	Gage
114	Cartridge profile failure (requiring more than 20 lbs. dead weight to insert in profile & alignment gage)		.25	3.3	See LI 9341551
115	Diameter of extractor groove, max, incorrect		.25	3.3	Gage

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TABLE II. QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 5		DRAWING NUMBER
4.4.2.1	Cartridge, 7.62MM, Match, XM852				9341551
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
116	Diameter of head, incorrect		.25	3.3	Gage
117	Thickness of head, incorrect		.25	3.3	Gage
118	Length to shoulder, incorrect		.25	3.3	Gage
119	Depth of primer, incorrect		.25	3.3	Gage
	Testing				
	Bullet extraction 3/	20		3.4	4.5.1
	Residual stress 3/	20		3.5	4.5.2
	Accuracy 4/	84		3.6	4.5.3
	Velocity 4/	20		3.7	4.5.4
	Chamber pressure 4/	20		3.8	4.5.5
	Port pressure 4/	20		3.9	4.5.6
	Function & Casualty 5/ (Rifle, 7.62MM, M14, National Match)	100		3.10	4.5.7
	Grain configuration 3/	10		3.11	4.5.8
	Hardness 4/	10		3.3	4.5.9

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TABLE II. QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63450 (AR)

PARAGRAPH	TITLE		SHEET 3 OF 5		DRAWING NUMBER 9341551
4.4.2.1	Cartridge, 7.62MM, Match, XM852				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Cartridge, discolored, dirty, oily, smeared 2/		1.5	3.3	Visual
202	Case dent 2/		1.5	3.3	Visual
203	Case draw scratch 2/		1.5	3.3	Visual
204	Case scratch 2/		1.5	3.3	Visual
205	Case scaly metal 2/		1.5	3.3	Visual
206	Case fold 2/		1.5	3.3	Visual
207	Case wrinkle 2/		1.5	3.3	Visual
208	Case buckle 2/		1.5	3.3	Visual
209	Case bulge 2/		1.5	3.3	Visual
210	Illegible or missing head stamp 2/		1.5	3.3	Visual
211	Defective head 2/		1.5	3.3	Visual
212	Defective mouth 2/		1.5	3.3	Visual
213	Bullet dent 2/		1.5	3.3	Visual
214	Bullet scratch 2/		1.5	3.3	Visual
215	Scaly metal (bullet) 2/		1.5	3.3	Visual
216	Nicked or dented primer 2/		1.5	3.3	Visual
217	No waterproofing material (primer pocket joint) 2/		1.5	3.3	Visual
218	Diameter of extractor groove, min, incorrect		1.5	3.3	Gage
219	Workmanship		2.5	3.12	Visual

NOTES:

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TABLE II. QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63450(AR)

PARAGRAPH	TITLE		SHEET 4 OF 5		DRAWING NUMBER 9341551
4.4.2.1	Cartridge, 7.62MM, Match, XM852				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	1/ Each light weight cartridge shall be disassembled and the propellant weighed. Each such cartridge found to contain 25 grains or more of propellant shall be classed as a major defect. Any cartridge containing less than 25 grains of propellant shall be classed as a critical defect. 2/ Refer to MIL-STD-636 for visual defects. In the event of conflict between paragraphs of this specification and MIL-STD-636 as to defect classification, the classification specified by specification shall apply. 3/ Failure of two or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If one cartridge fails in the first test, a second sample consisting of double the number of cartridges in the first sample shall be tested. If any failing cartridges are found in the second sample, the lot shall be rejected. 4/ Failure of the cartridges to comply with the applicable requirement shall be cause for rejection of the lot subject to testing of a second sample consisting of double the quantity of cartridges used in the first test. Failure of the cartridges in the second sample to comply with the applicable requirement shall be cause for rejection of the lot.				

NOTES:

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TABLE II. QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63450(AR)

PARAGRAPH	TITLE		SHEET 5 OF 5		DRAWING NUMBER 9341551
4.4.2.1	Cartridge, 7.62MM, Match, XM852				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	5/ The lot shall be rejected when function and casualty defects, plus firing defects observed in all other firing tests exceed the acceptance number for the cumulative sample in Table III. If the number of defects found in the first test exceeds the acceptance number for the first sample, but is equal to or less than the acceptance number for the cumulative sample, a second sample consisting of double the quantity specified under function and casualty test shall be fired in the service weapon specified therefore. If the total number of defects in the combined first and second samples exceeds the acceptance number for the cumulative sample, the lot shall be rejected. If in testing a second sample, defects other than those for which the second sample is being tested should occur to the extent that they exceed the acceptance number for the cumulative sample, the lot shall be rejected.				

NOTES:

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection Tables.

4.4.3.1 Packaging, packing, and marking. Inspection for packaging, packing and marking shall be in accordance with MIL-STD-644. During or immediately prior to the packaging operation, 100 percent examination of the cartridges shall be performed to ascertain that the cartridge type conforms to the drawing. Occurrence of a high pressure test cartridge shall be classed as a critical defect. Occurrence of any incorrect type other than high pressure test shall be classed as a major defect. All non-conforming cartridges shall be rejected.

4.4.3.2 Firing defects. Firing defects and acceptance numbers shall be as specified in Table III.

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TABLE III

Defects	Acceptance Number: First Sample	Acceptance Number: Cumulative (First and Second Sample)
1. Misfire	0	1
2. Bullet remaining in bore 1/	0	-
3. Primer leak		
a. Perforation in firing pin indent in primer cup	1	2
b. Escape of gas through primer cup other than 3a	1	2
c. Escape of gas around primer cup 2/	3	5
d. Primer remains in pocket, but is physically loose	1	2
e. Blown primer or primer falls out of pocket or retraction of bolt	0	1
4. Case casualties		
a. Longitudinal split 3/		
(1) Neck and shoulder (I & S)	5	9
(2) Body (J)	1	2
(3) Body (K)	0	1
(4) To head (L)	0	1
(5) Through head (M)	0	1
b. Circumferential rupture 3/		
(1) Partial, shoulder or body (J and S)	1	2
(2) Partial, body (K)	0	1
(3) Partial, head (L)	0	1
(4) Complete	0	1
5. Failure to Extract	0	1

1/ No second sample permitted. Lot shall be rejected.
2/ Gas escapes around more than 50 percent of periphery of cup.
3/ For location of defects indicated by letters in parenthesis, see Drawing C7643674.

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4.4.4 Inspection equipment. The inspection equipment required to perform the examination and test prescribed herein is described in the Paragraph Reference/Inspection method column in the tables starting with Paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.2 herein.

4.4.4.1 Index of inspection equipment list. The examination tests shall be made using equipment listed on LI 9341551.

4.5 Test methods and procedures.

4.5.1 Bullet extraction. The test shall be conducted in accordance with SCATP 7.62MM.

4.5.2 Residual stress (Mercurous Nitrate). The test shall be conducted in accordance with SCATP 7.62MM.

4.5.3 Accuracy. The test shall be conducted in accordance with SCATP 7.62MM.

4.5.4 Velocity. The test shall be conducted in accordance with SCATP 7.62MM.

4.5.5 Chamber pressure. The test shall be conducted in accordance with procedures referenced in LI 9341551.

4.5.6 Port pressure. The test shall be conducted in accordance with SCATP 7.62MM.

4.5.7 Function and casualty. The test shall be conducted in accordance with SCATP 7.62MM, except that all firings shall be semi-automatic. Two rifles shall be used, and one-half of the cartridges shall be fired in each weapon.

4.5.8 Grain configuration. The test samples shall be prepared and evaluated in accordance with MIL-STD-1170.

4.5.9 Hardness testing. The bullets shall be extracted, the propellant removed and the primers extracted. Each cartridge case of the sample shall be prepared and placed on the appropriate test fixture for testing in accordance with Federal Test Method Standard No. 151, Method 244.1. The hardness values of the sample cases for each prescribed point shall be computed and charted in accordance with the drawing requirements.

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4.5.10 Defect penalty. In any ballistic test, except function and casualty, in which the occurrence of a firing defect listed in Table III prevents the obtaining of a reliable result for the characteristic being tested, an additional shot shall be fired. That particular test shall not be penalized, but the total ballistic sample shall be penalized for such defects in accordance with Table III.

4.5.11 Test validity. If for any reason the proving ground considers that the test conditions have detrimentally affected the test results, the test shall be declared invalid and a new test shall be performed with additional samples.

5. PREPARATION FOR DELIVERY

5.1 Packing. Level A (World Wide Shipment). The cartridges shall be packed in accordance with Drawing 9341748, 9341749 and 9341750.

5.2 Marking and labeling. Packing boxes shall be marked and labeled in accordance with the applicable drawings cited in 5.1.

6. NOTES

6.1 Ordering data. See MIL-A-48078.

6.2 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs as required to Commander, ARRADCOM, Dover, NJ 07801, ATTN: DRDAR-QAA-I.

6.3 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.4 Provisions for the supply, maintenance and disposition of mandatory ballistic test equipment for acceptance inspection purposes.

6.5 Provisions for the submission of acceptance inspection reports containing final inspection results for each lot of ammunition presented to the Government.

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6.6 Provisions for the identification and submission of grain configuration photomicrographs prepared in accordance with MIL-STD-1170 where such submission is required.

6.7 Head stamp. Designation "NM" (abbreviation of National Match will be applied to cartridges in lot(s) specifically manufactured for Camp Perry National Matches.

Custodian:

Army-AR

Preparing Activity:

Army-AR

Project No. 1305-A870

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